

QUESTION 1: HOUGH TRANSFORM IN NOISY IMAGES

Scenario

A system uses Hough Transform to detect lines in noisy images but produces inaccurate results. Analyze the effect of noise on detection and suggest improvements based on preprocessing or parameter selection.

Answer

1. Problem Analysis

The **Hough Transform** is a technique used to detect geometric shapes such as lines and circles. It works by converting edge points from an image into a parameter space. When the input image contains noise, unwanted pixels and false edges are also detected.

These noisy points can generate incorrect votes in the Hough parameter space. As a result, the system may detect false or inaccurate lines.

2. Effect of Noise

Noise can cause:

- False edge points to appear.
- Incorrect lines to be detected.
- Extra lines to be generated.
- Weak or fragmented lines to be detected.
- Reduced detection accuracy.

3. Recommended Improvements

Preprocessing:

Apply **Gaussian filtering** or **Median filtering** to reduce noise before detecting edges.

Edge Detection:

Use **Canny Edge Detection** to obtain cleaner and stronger edges.

Parameter Selection:

The following Hough parameters should be properly adjusted:

- **Threshold:** Controls the minimum number of votes required to detect a line.
- **Minimum Line Length:** Removes very short unwanted lines.
- **Maximum Line Gap:** Controls the maximum gap allowed between connected line segments.

4. Conclusion

Noise significantly affects Hough Transform because unwanted pixels can generate false votes. Therefore, **noise removal + Canny edge detection + proper Hough parameter selection** can improve line detection accuracy.

QUESTION 2: PCA FOR FEATURE OPTIMIZATION

Scenario

A system extracts a large number of features, leading to high computational cost and redundancy. Analyze how Principal Component Analysis helps in reducing dimensionality and improving efficiency.

Answer

1. Problem Analysis

In computer vision, an image can generate a large number of features. Processing all these features requires more memory, storage, and computational time. Some features may also contain redundant or similar information.

Principal Component Analysis (PCA) is used to reduce the dimensionality of the feature data while preserving the most important information.

2. Working of PCA

PCA transforms the original features into a smaller number of new features called **principal components**.

The principal components are arranged according to the amount of information or variance they contain. The most important components are retained, while less important components can be removed.

Example:

100 original features → PCA → 20 important components

This significantly reduces the amount of data that needs to be processed.

3. Advantages of PCA

- Reduces dimensionality.
- Removes redundant information.
- Reduces computational complexity.
- Reduces memory requirements.
- Speeds up feature processing.
- Provides a compact feature representation.
- Can improve the efficiency of classification and matching.

4. Conclusion

PCA improves feature optimization by retaining the most informative components and removing unnecessary dimensions. Hence, it makes computer vision systems **faster, more efficient, and less computationally expensive**.

QUESTION 3: EDGE DETECTION IN LOW CONTRAST IMAGES

Scenario

An image has low contrast, causing edge detection algorithms to fail. Analyze the problem and explain how preprocessing enhances edge detection results.

Answer

1. Problem Analysis

Edge detection algorithms such as **Sobel and Canny** identify boundaries based on changes in pixel intensity.

In a low-contrast image, the difference between the object and its background is very small. Therefore, the intensity gradient is weak, making it difficult for the edge detection algorithm to identify important boundaries.

2. Problems Caused by Low Contrast

- Important edges may not be detected.
- Detected edges may be very weak.
- Edges may become fragmented.
- Object boundaries may not be clear.
- Edge detection accuracy decreases.

3. Preprocessing Techniques

Histogram Equalization:

Improves the overall contrast by redistributing the intensity values.

CLAHE:

Contrast Limited Adaptive Histogram Equalization improves contrast locally and is useful when different regions have different illumination levels.

Noise Reduction:

Gaussian or Median filtering can be applied to remove noise before edge detection.

4. Recommended Processing Flow

Low-Contrast Image

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Contrast Enhancement

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Noise Reduction

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Canny/Sobel Edge Detection

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Improved Edge Map

5. Conclusion

Preprocessing increases the intensity difference between important regions and their surroundings. Therefore, applying **contrast enhancement and noise reduction before edge detection** produces stronger, clearer, and more continuous edges.

QUESTION 4: CORNER DETECTION IN TEXTURED REGIONS

Scenario

A system applies corner detection in highly textured regions, resulting in excessive keypoints. Analyze the issue and suggest methods to improve feature selection.

Answer

1. Problem Analysis

Highly textured regions contain many intensity changes in different directions. Corner detection algorithms such as **Harris Corner Detection** may therefore identify a very large number of keypoints.

Although these keypoints provide information, detecting too many of them can reduce system efficiency.

2. Problems Caused by Excessive Keypoints

- Increases computational cost.
- Increases feature matching time.
- Produces redundant keypoints.
- May include weak or unstable features.
- Can make matching more difficult.

3. Methods to Improve Feature Selection

Response Threshold:

Increase the threshold so that only strong corners are selected.

Non-Maximum Suppression (NMS):

Retain only the strongest corner within a local region.

Minimum Distance:

Set a minimum distance between keypoints to prevent multiple nearby points from being selected.

Keypoint Ranking:

Rank keypoints according to their corner response and select the strongest ones.

Maximum Keypoint Limit:

Set a maximum number of keypoints to control computational complexity.

4. Conclusion

Excessive keypoints in textured regions reduce efficiency and may introduce redundant information. Therefore, **thresholding, Non-Maximum Suppression, minimum-distance constraints, and keypoint ranking** can improve feature selection while retaining useful features.

QUESTION 5: COMBINED FEATURE DETECTION AND MATCHING

Scenario

A system detects features correctly but fails during matching due to incorrect descriptor selection. Analyze the issue and explain how appropriate feature descriptors improve matching performance.

Answer

1. Problem Analysis

Feature detection and feature description are different stages of a computer vision system.

- **Feature Detection:** Identifies important points or keypoints.
- **Feature Description:** Represents the local appearance around each keypoint.
- **Feature Matching:** Compares descriptors between two images to find corresponding points.

Therefore, correct feature detection alone does not guarantee successful matching.

2. Why Matching Fails

Matching can fail when the selected descriptor:

- Is sensitive to scale changes.
- Is sensitive to rotation.
- Is affected by illumination changes.
- Cannot distinguish similar image regions.
- Is unsuitable for the detected features.

3. Comparison of Feature Descriptors

	Descriptor Main Advantage	Suitable Application
SIFT	Robust to scale and rotation	Accurate feature matching
ORB	Fast and computationally efficient	Real-time applications
SURF	Robust to scale and rotation	Robust feature matching

4. SIFT

SIFT (Scale-Invariant Feature Transform) is suitable when images have changes in scale or orientation. It provides robust descriptors for accurate feature matching.

5. ORB

ORB (Oriented FAST and Rotated BRIEF) is computationally faster than SIFT and is suitable for applications where real-time performance is important.

6. SURF

SURF (Speeded-Up Robust Features) provides robustness to scale and rotation changes and can be used when reliable matching is required.

7. Conclusion

Feature detection alone is not sufficient for successful image matching. The descriptor should be selected according to the image conditions and expected transformations. **SIFT is preferred for robust and accurate matching, while ORB is preferred when speed and computational efficiency are more important.**